Recommended Datasets

1. **Credit Card Fraud Detection (Kaggle)
    - Link: https://www.kaggle.com/datasets/mlg-ulb/creditcardfraud Kaggle
    - Description: Classic credit card transactions dataset, with fraud labels.
    - Use Case: Good for initial model training, classification, imbalance handling.
2. **Fraud Detection Dataset (Kaggle)
    - Link: https://www.kaggle.com/datasets/goyaladi/fraud-detection-dataset Kaggle
    - Description: Larger set of transaction data, various features, fraud labels.
    - Use Case: More variety of features, optional for richer feature engineering.
3. **Online Payments Fraud Detection Dataset (Kaggle)
    - Link: https://www.kaggle.com/datasets/rupakroy/online-payments-fraud-detection-dataset Kaggle
    - Description: Focused on online payments, useful for e-wallet/fintech style transaction fraud.
    - Use Case: Matches your mobile/online payment scope well (you referenced UPI/wallet style).
4. **Bank Account Fraud Dataset Suite (NeurIPS 2022) (Kaggle)
    - Link: https://www.kaggle.com/datasets/sgpjesus/bank-account-fraud-dataset-neurips-2022 Kaggle
    - Description: Synthetic bank-account based fraud datasets, multiple sub-sets.
    - Use Case: Good for modelling bank-account & transaction behavior beyond card only.
5. **Synthetic Mobile Money Transaction Dataset (Mendeley)
    - Link: https://data.mendeley.com/datasets/zhj366m53p/1 Mendeley Data
    - Description: Synthetic mobile money transaction records, designed for fraud modelling.
    - Use Case: Very relevant for mobile-money/fintech use-case (you referenced UPI/wallet).

Suggestions for Use in Your Project

- Choose **one primary** dataset (e.g., Credit Card Fraud or Online Payments).
- Optionally **merge** or **augment** with a second dataset if feature richness is needed (e.g., Bank Account suite or Mobile Money).
- Ensure dataset includes or derive fields that match your dataset specification (Section 13): `transaction_id`, `user_id`, `amount`, `timestamp`, `device_id`, `location`, `merchant_type`, `payment_method`, `label`, etc.
- For missing fields (e.g., `device_id`, `location`) you may synthesize them or derive new fields (device change flag, location deviation) to match your architecture.
- Handle class imbalance (fraud vs non-fraud) as specified in preprocessing section

#	Name	Source & Link	Description & Key Features	Suitability Notes
1	Credit Card Fraud Detection	Kaggle — Credit Card Fraud Detection (Kaggle)	Classic dataset: anonymised credit-card transactions (~284,000 rows), fraud labelled.	Good base dataset for supervised fraud detection; may need additional fields (device_id, location) added for full spec.
2	Fraud Detection Dataset	Kaggle — Fraud Detection Dataset (Kaggle)	Larger variety transactions; labelled fraud/non-fraud.	Useful for richer feature engineering; still may need device/location fields.
3	Online Payments Fraud Detection	Kaggle — Online Payments Fraud Detection Dataset (Kaggle)	Specifically online payments dataset (fintech/UPI style).	Closely aligned with mobile/online wallet scope; good fit for your system.
4	Bank Account Fraud Dataset Suite (NeurIPS 2022)	Kaggle — Bank Account Fraud Dataset Suite (Kaggle)	Synthetic bank-account data, multiple sub-sets, with controlled biases.	Useful for bank side fraud modelling; good supplementary data.
5	Fraud Detection Transactions Dataset	Kaggle — Fraud Detection Transactions Dataset (Kaggle)	More general transaction dataset for fraud modelling.	Can provide additional variety / help with feature engineering.
6	Fraudulent E-Commerce Transactions	Kaggle — Fraudulent E-Commerce Transactions (Kaggle)	Synthetic dataset, e-commerce focus, includes online transaction features.	Good for online commerce angle; may require adaptation for banking/UPI context.
7	Financial Fraud Detection (Mobile Money)	Kaggle — Financial Fraud Detection Dataset (Kaggle)	Synthetic mobile money transactions for fraud research.	Very relevant to your mobile wallet/UPI scenario; excellent fit for dataset specification.

#	Dataset Name	Source (Downloadable)	Summary of Contents
1	**Credit Card Fraud Detection**	Kaggle (Kaggle)	~284,000 anonymised credit card transactions labelled fraud or not.
2	**Fraud Detection Dataset**	Kaggle (Kaggle)	Larger variety of transaction types and features; fraud labels included.
3	**Online Payments Fraud Detection Dataset**	Kaggle (Kaggle)	Online payments / fintech style transactions, with fraud labels.

#	Dataset Name	Source (Downloadable)	Summary of Contents
4	**Bank Account Fraud Dataset Suite (NeurIPS 2022)**	Kaggle (Kaggle)	Synthetic bank account transaction datasets, multiple subsets, fraud-related.
5	**Fraud Detection Transactions Dataset**	Kaggle (Kaggle)	Broad transaction dataset aimed at fraud detection modelling.
6	**Fraudulent E-Commerce Transactions**	Kaggle (Kaggle)	E-commerce transactions (synthetic) with fraud labels; useful for online commerce component.
7	**Financial Fraud Detection (Mobile Money)**	Kaggle (Kaggle)	Mobile-money style transactions, fraud labels; good match for your UPI/Wallet context.